

# Lecture 2

Monday PM, 7/13

## Topics/outline

- Circular motion in magnetic fields
- Weak focusing
- Strong focusing
- Equations of motion/betatron oscillations

This lecture: learn how to describe the transverse motion (perpendicular to the primary/longitudinal motion)

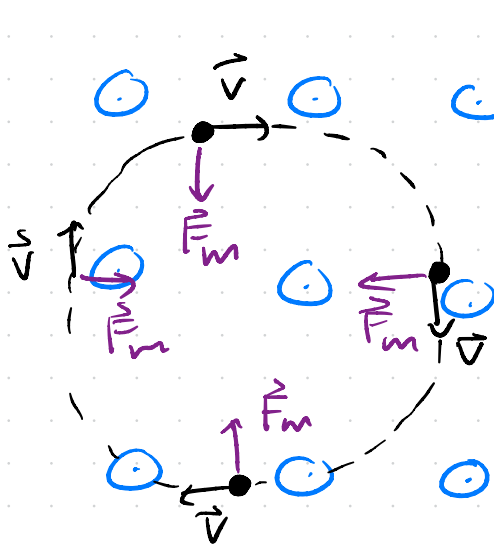
## Circular motion

Begin with the Lorentz force equation:

$$\vec{F} = \frac{d\vec{p}}{dt} = q(\vec{E} + \vec{v} \times \vec{B})$$

\* includes sign!   
 ↑ acceleration   
 ↑ steering / controlling trajectories

Simple example: charged particle moving perpendicular to uniform field



$\vec{B}$  (out of page)

Velocity is always  $\perp$  to magnetic field

$\Rightarrow$  circular orbit

Radius of orbit: central force  
\* non-relativistic, not accelerating

$$\vec{F}_m = q(\vec{v} \times \vec{B}) \Rightarrow F_m = qvB = \frac{mv^2}{R}$$

Circular motion

$$\Rightarrow R = \frac{mv}{qB}$$

- Context of a circular accelerator: use bending radius  $\rho$

$$\rho = \frac{P}{qB}$$

- Define the "rigidity" to be

$$B\rho = \frac{P}{q}$$

\*  $q = \pm e$  for proton/electrons  
 $Ze$  for ions

Dimensions [T.m]

- Energy is not changing, only direction

\* Recall: static magnetic fields do no work, cannot change the energy

Why? Increment of work  $dW$ :

$$dW = \vec{F} \cdot d\vec{\ell} \quad (\text{where } d\vec{\ell} = \vec{v} dt)$$

$$dW = q(\vec{v} \times \vec{B}) \cdot \vec{v} dt \Rightarrow (\vec{v} \times \vec{B}) \cdot \vec{v} = 0$$

always!

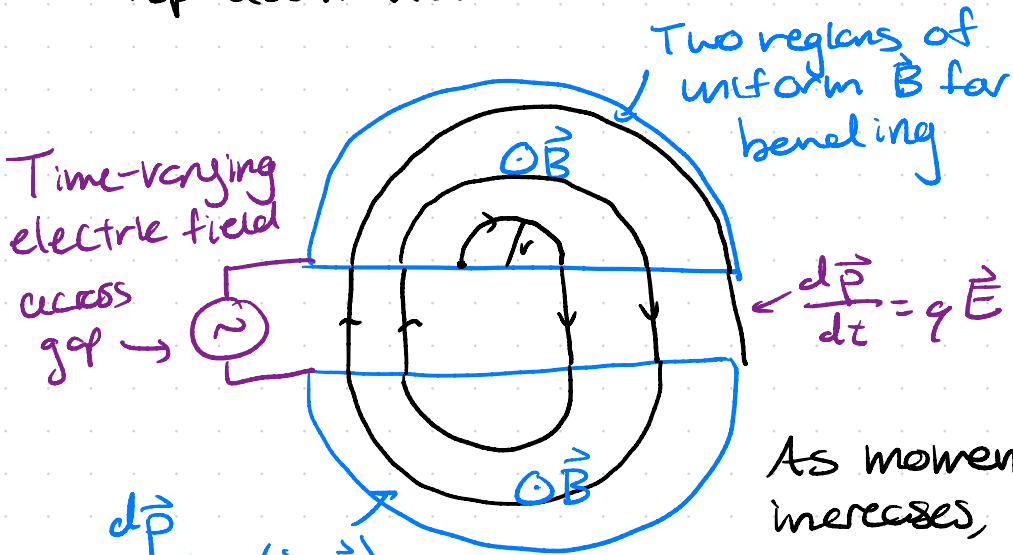
Consider two early accelerators: the cyclotron and the betatron

↓  
acceleration  
via applied  $\vec{E}$

↓  
induction-based  
acceleration

## Cyclotron:

Top-down view:



$$\frac{d\vec{p}}{dt} = q(\vec{v} \times \vec{B})$$

$$\frac{d\vec{p}}{dt} = q\vec{E}$$

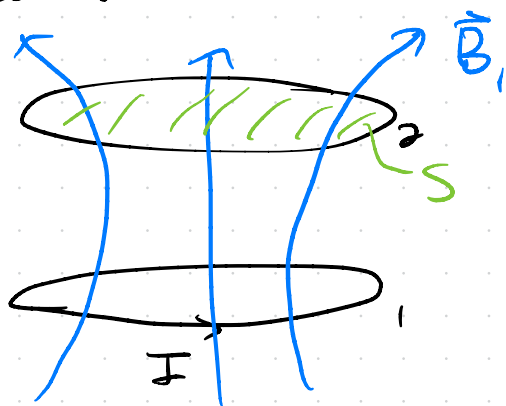
As momentum increases, so does  $\rho = \frac{p}{qB}$

↓  
Later: important in the motion of an off-momentum particle



## Betatron:

Uses time-varying  $\vec{B}$  fields to accelerate particles via induction



Flux through loop 2:

$$\Phi_B = \int_S \vec{B}_1 \cdot d\vec{A}$$

• Recall:  $\mathcal{E} = -\frac{d\Phi_B}{dt}$  electromotive force (emf)  
emf acts on charge carriers in the loop

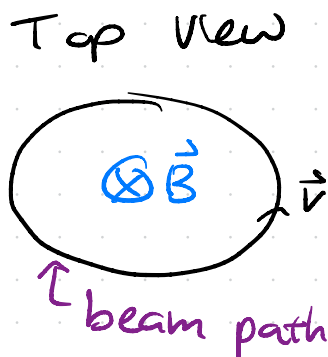
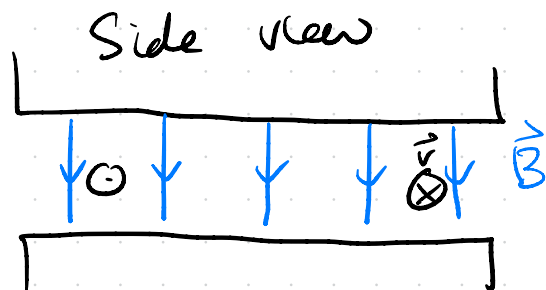
• Sign?  $\Rightarrow$  Lenz's Law "nature abhors a change in flux" induced current flows in direction to oppose change

Faraday's Law:

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A}$$

Differential form:  $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

In a betatron, beam is loop 2

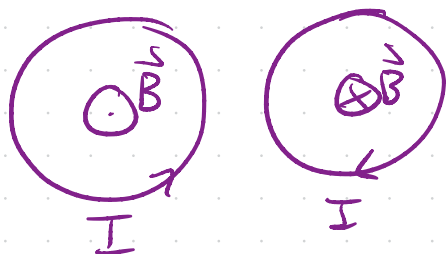


Start to increase  $\vec{B}$  strength

\* Concept check: Which direction is the induced emf?

Lenz's Law: Oppose change,  $\vec{B}$  increasing into the page, field from induced current should point out of the page  
 $\Rightarrow$  emf is CCW, accelerates beam

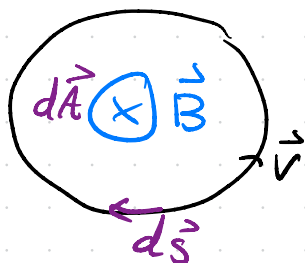
\* Reminder:



Use right hand rule

Check mathematically need to establish direction of  $d\vec{A}$ ,  $d\vec{s}$ , again related by RHR

Choose  $d\vec{A}$  is along  $\vec{B}$  into page



$$\vec{B} \uparrow \Rightarrow \frac{d\Phi}{dt} > 0$$

$$\oint \vec{E} \cdot d\vec{s} = - \frac{d}{dt} \underbrace{\int \vec{B} \cdot d\vec{A}}_{>0}$$

$\Rightarrow \oint \vec{E} \cdot d\vec{s} < 0$   $\vec{E}$  opposite  $d\vec{s}$ , along  $\vec{v}$   
particles accelerated along their orbit

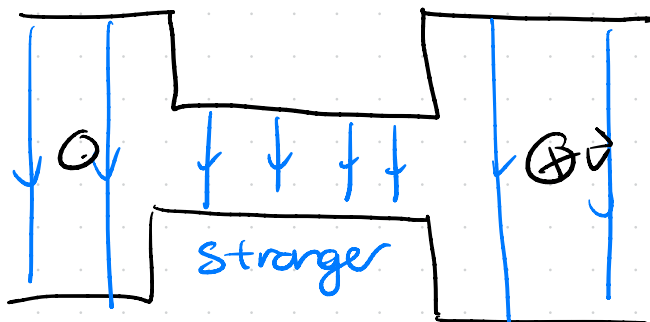
\* Recall: established  $p \uparrow$  as  $p \uparrow$  for some  $B$

$B$  is now changing, want to maintain fixed  $p$  (unlike cyclotron)

- Circular orbit depends on  $\vec{B}$  at beam
- Acceleration depends on  $\frac{\partial \vec{B}}{\partial t}$  enclosed

Balance forces (H.W. 3.1 has full proof)

$\Rightarrow$  Find that  $B_{\perp}(\rho) = \frac{1}{2} \overline{B_{\perp}}(\rho)$  widened  $\frac{1}{2}$  criterion  
Field  $\perp$  to  $\vec{v}$  average within orbit



provides stable orbit + acceleration

## Weak focusing

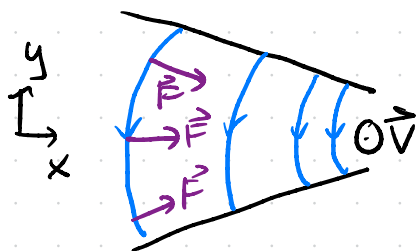
What happens when  $\vec{v}$  is not totally  $\perp$  to  $\vec{B}$ ?



"Guided" by field line  
with no vertical restoring  
force, eventually lost

$\hookrightarrow$  Requires vertical  
focusing

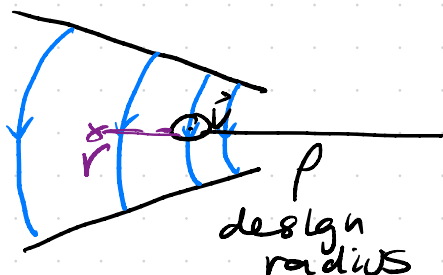
Add some  $B_x \neq 0$ , particles above or below the midplane feel a vertical force



BUT  $\vec{\nabla} \times \vec{B} = 0$

$$\Rightarrow \frac{\partial B_x}{\partial y} - \frac{\partial B_y}{\partial x} = 0$$

Non-zero  $\frac{\partial B_x}{\partial y}$  requires  $\frac{\partial B_y}{\partial x} \Rightarrow$  field gets weaker at larger radius, decreasing in  $x$



Beam radius is

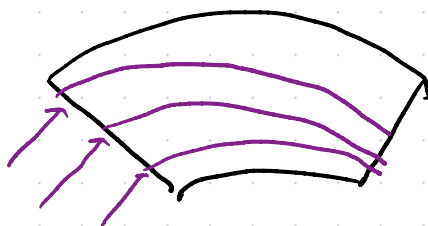
$$\propto B^{-1}$$

$\Rightarrow$  larger  $r$ , weaker  $\vec{B}$

BUT this is counteracted by focusing in the bend direction based on path length

$\hookrightarrow$  larger  $r \rightarrow$  longer path through field  $\rightarrow$  greater deflection

Will see in sector bend magnet



- Can outweigh defocusing from  $\frac{\partial B_y}{\partial x}$
- Resulting transverse motion takes the form of oscillations around the ideal closed orbit  $\Rightarrow$  "betatron oscillations" of the form

$$\ddot{x} + \omega_x^2 x = 0$$

$$\ddot{y} + \omega_y^2 y = 0$$

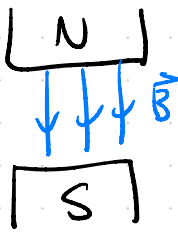
(See H.W. 3.1-3.2)  
S.Y. Lee 1.I.3

- Limitation to field gradients while maintaining stable orbits  $\Rightarrow$  "weak focusing"
- Not enough for high energy accelerators, oscillation in radial direction becomes impractically large

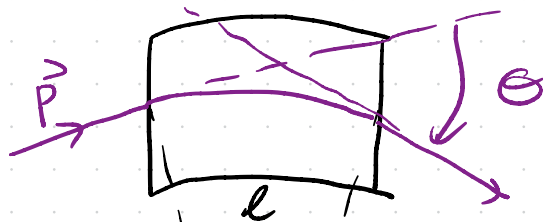
# Strong focusing

- Two separate magnets
  - Dipole to bend beam
  - Quadrupole to focus beam
- Dipole: follows from previous discussion, but over a limited region

Side view



Top view



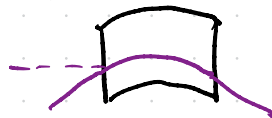
Deflects beam  
by angle  $\theta$  from  
original trajectory

$$\theta = \frac{l}{\rho}$$

Note: different types, different edge effects (more later)



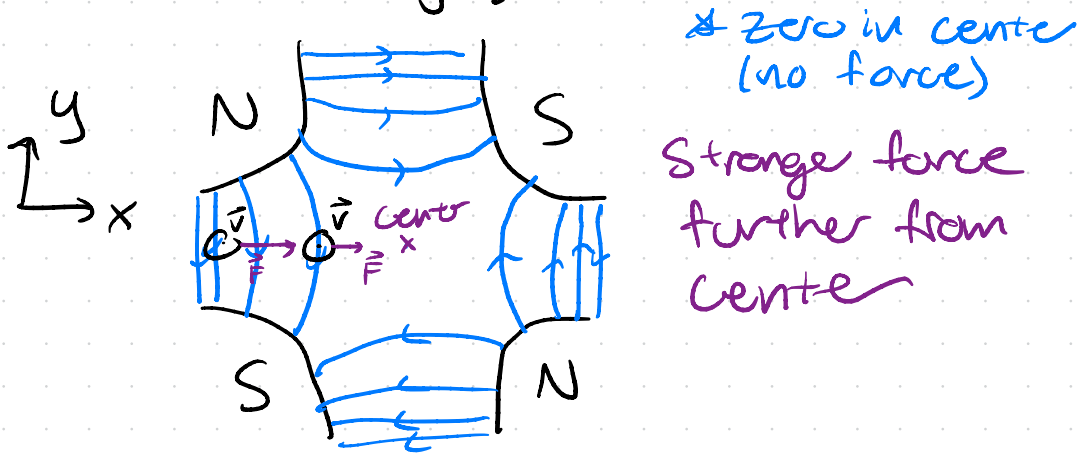
"sector"  
bend



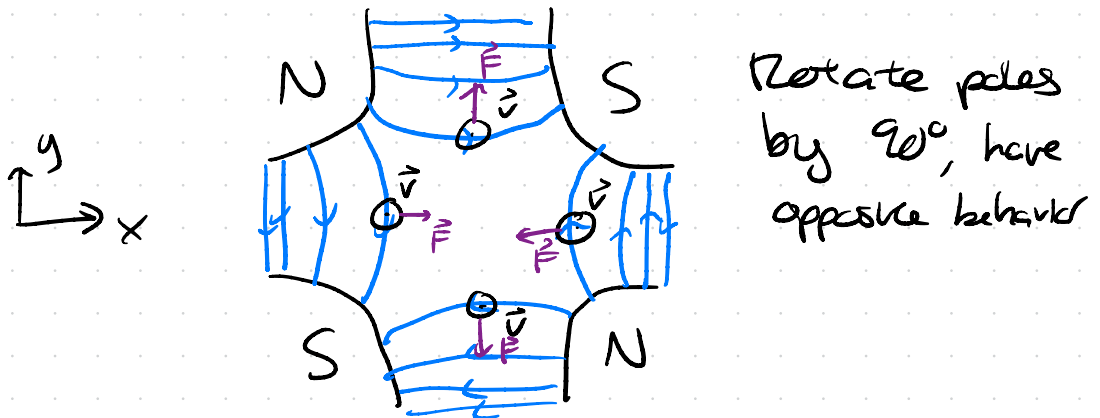
"rectangular"  
bend

# Quadrupole magnet

- Take gradient idea from betatron (focus in one plane), alternate
- Achieved using quadrupole



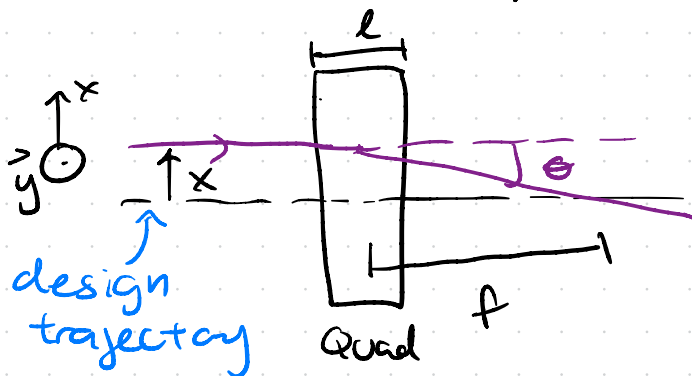
This arrangement focuses in x, but defocuses in y:



Alternate series of quadrupoles for focusing  
 $\Rightarrow$  more tomorrow



Look from the top or side: deflection



Use the language of geometric optics: focal length, paraxial trajectory

$$\theta = \frac{-l}{\rho} = -l \left( \frac{q B_y}{P} \right) = \frac{-x}{f} \quad \text{"thin" lens} \quad f \gg l$$

For some general  $\vec{B} = B_x \hat{x} + B_y \hat{y}$  (assuming  $B_z = 0$ )

$$\text{Have } \vec{\nabla} \cdot \vec{B} = 0 \Rightarrow \frac{\partial B_x}{\partial x} = 0 = \frac{\partial B_y}{\partial y}$$

$$\vec{\nabla} \times \vec{B} = 0 \Rightarrow \frac{\partial B_x}{\partial y} = \frac{\partial B_y}{\partial x}$$

Quadrupole field has simple form

$$B_x(y) = B' y \quad B_y(x) = B' x$$

Fields vary linearly with coordinate with constant gradient

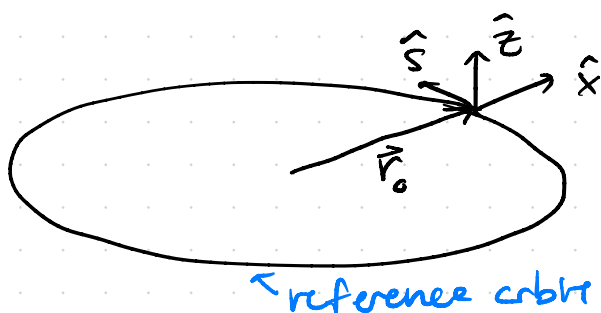
$$\theta = \frac{+l q B' x}{P} = \frac{x}{f} \quad \left[ \frac{1}{f} = \frac{l q B'}{P} \right] \quad \text{larger } B' \text{ shorter } f$$

# Equations of motion

Now that we have an intro to our linear magnets, let's work on finding transverse equations of motion

Need to introduce curvilinear (Frenet-Serret) coordinates

\*Note: lots of different conventions, here I follow S.Y. Lee (Section 2.I.1)



$\hat{s}$  along orbit  
 $s$  is coordinate along reference orbit

$x$ : horizontal displacement  
 $z$ : vertical displacement } from reference orbit

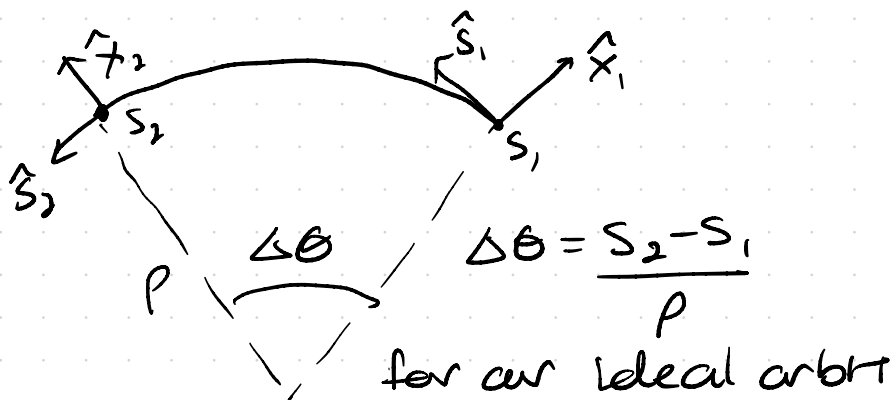
Any position can be expressed by

$$\vec{r} = \vec{r}_0 + x\hat{x} + z\hat{z} \quad \text{where} \quad \hat{s}(s) = \frac{d\vec{r}_0(s)}{ds},$$

$$\hat{x}(s) = -\rho(s) \frac{d\hat{s}(s)}{ds}, \quad \hat{z}(s) = \hat{x}(s) \times \hat{s}(s)$$

radius of curvature, constant for ideal orbit

Note all functions of  $S$ :



I will use prime  $'$  for differentiation w.r.t.  $S$  and dot  $\dot{\phantom{x}}$  for w.r.t. time

### Finding EOM

- Newton's laws (today + homework)
- Hamiltonian description
  - ↳ see S.Y. Lee Sec. 2. II. 1-3
  - H.W. Sec. 5.4

Start from Lorentz force  
(neglecting acceleration for now)

$$\vec{F} = \frac{d\vec{p}}{dt} = q(\vec{v} \times \vec{B}) \quad \frac{d\rho}{dt} = \frac{d}{dt}(x m \vec{v})$$

$$\Rightarrow \gamma m \ddot{\vec{r}} = q(\vec{v} \times \vec{B})$$

$$\vec{v} = v_x \hat{x} + v_y \hat{y} + v_z \hat{z}$$

↑ note: no assumption of  $\vec{B}$  (dipole, quadrupole, etc.)

With  $\vec{r} = \vec{r}_0 + x \hat{x} + z \hat{z}$ , for deviations around the ideal orbit this is

$$\vec{r} = (\rho + x) \hat{x} + z \hat{z}$$

We get (your homework)

$$B\rho = \frac{p}{q} \tau$$

$$\ddot{x} - (\rho + x) \dot{\theta}^2 = \frac{v_s^2 B_z}{B\rho}, \quad \ddot{z} = -\frac{v_s^2 B_x}{(B\rho)} \quad \text{rigidity} \downarrow$$

\* two assumptions: No  $B_z$  (longitudinal B field)

$$v_x \ll v_s \text{ and } v_z \ll v_s \Rightarrow p \approx \gamma m v_s$$

Oscillations in time

Want to instead describe motion as  $s$  evolves

Use chain rule (your homework)

$$dt \rightarrow ds$$

$$\Rightarrow \left[ \begin{aligned} x'' - \frac{p+x}{\rho^2} &= \pm \frac{B_z}{B\rho} \left(1 + \frac{x}{\rho}\right)^2 \\ z'' &= \mp \frac{B_x}{B\rho} \left(1 + \frac{x}{\rho}\right)^2 \end{aligned} \right]$$

positive charge  
negative charge  
positive charge  
negative charge

- Note so far these equations describe a particle with ideal momentum  $p=p_0$
- "off-momentum" particles have additional factor of  $p_0/p$  (more later this week)

Next up: incorporate  $B_z$  and  $B_x$  to describe motion through magnets

First: dipole + quadrupole

↳ Field will have the form

$$B_z = \mp B_0 + \frac{\partial B_z}{\partial x} x = \mp B_0 + B_1 x$$

sign gives  
CCW orbit  
along  $\hat{s}$

↑  
bending term

↑  
quadrupole

$$B_x = \frac{\partial B_z}{\partial x} z = B_1 z$$

$\hat{=}$  quadrupole

$$\frac{\partial B_z}{\partial x} = \frac{\partial B_x}{\partial z}$$

Notes:

- No bending term in  $x$  (planar acceleration would add vertical bend)

$$\frac{\partial B_x}{\partial x} = \frac{\partial B_z}{\partial z} = 0$$

These terms would couple two transverse coordinates, only want to consider uncoupled betatron oscillations for now

- $B_1 = \frac{\partial B_z}{\partial x}$  is the gradient

Plug in and rearrange, keep only linear terms in coordinate (for now):

+ taking + charge case for simplicity

$$x'' - \frac{1}{\rho} - \frac{x}{\rho^2} = \frac{B_z}{B\rho} \left( 1 + \frac{2x}{\rho} + \frac{x^2}{\rho^2} \right)$$

$$B_z = -B_0 + B_1 x$$

$$\Rightarrow x'' - \frac{1}{\rho} - \frac{x}{\rho^2} = -\frac{B_0}{B\rho} \left(1 + \frac{2x}{\rho}\right) + \frac{B_1 x}{B\rho} \left(1 + \frac{2x}{\rho}\right)$$

neglect 2nd order in x

$$\frac{B_0}{B\rho} = \frac{1}{\rho} \quad \text{dipole bending for closed orbit, on-momentum um}$$

$$x'' - \cancel{\frac{1}{\rho}} - \frac{x}{\rho^2} = -\cancel{\frac{1}{\rho}} - \frac{2x}{\rho^2} + \frac{B_1 x}{B\rho}$$

$$\boxed{x'' + x \left( \frac{1}{\rho^2} - \frac{B_1}{B\rho} \right) = 0}$$

EOM for this term of the field, linear terms

Form of a Hill equation

$$x'' + K_x(s)x = 0 \quad z'' + K_z(s)z = 0$$

Oscillations where "spring constant" depends on s

Plane, no vertical bending

$$K_x(s) = \frac{1}{\rho^2} + K_1(s), \quad K_z(s) = -K_1(s)$$

Bending term

Focusing term

$$K_1(s) = \mp \frac{B_1(s)}{B\rho}$$

Remember  $B_1 \rightarrow$  gradient  
 $\frac{\partial B_z}{\partial x} = \frac{\partial B_x}{\partial z}$

Following S.Y. Lee notation

Sign convention:  $K_1 > 0$  horizontal focusing

• Strong vs weak focusing:

↳ define focusing index  $n(s)$

$$n(s) = \pm \rho^2 K_1(s)$$

Weak focusing:  $0 \leq n \leq 1$

Strong focusing:  $|n| \gg 1$

• Single-function magnets:

Bending dipole:  $K_x = \frac{1}{\rho^2}$   $K_z = 0$

Quadrupole:  $K_x = K_1 = -K_z$

Focusing in one plane, defocusing  
in the other

Tomorrow: periodic lattice